

FARMMANAGEMENTSYSTEM

A Salesforce-Based Application for Digitizing Agricultural Operations

Project Documentation

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UserStory:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

Purpose:

The main purpose of this project is to:

- Digitize agricultural data management
- Improved data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

Problem Statements – Farm Management System

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1		Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	
PS-2	 Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient 
PS-3		Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied 
PS-4	 System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure 

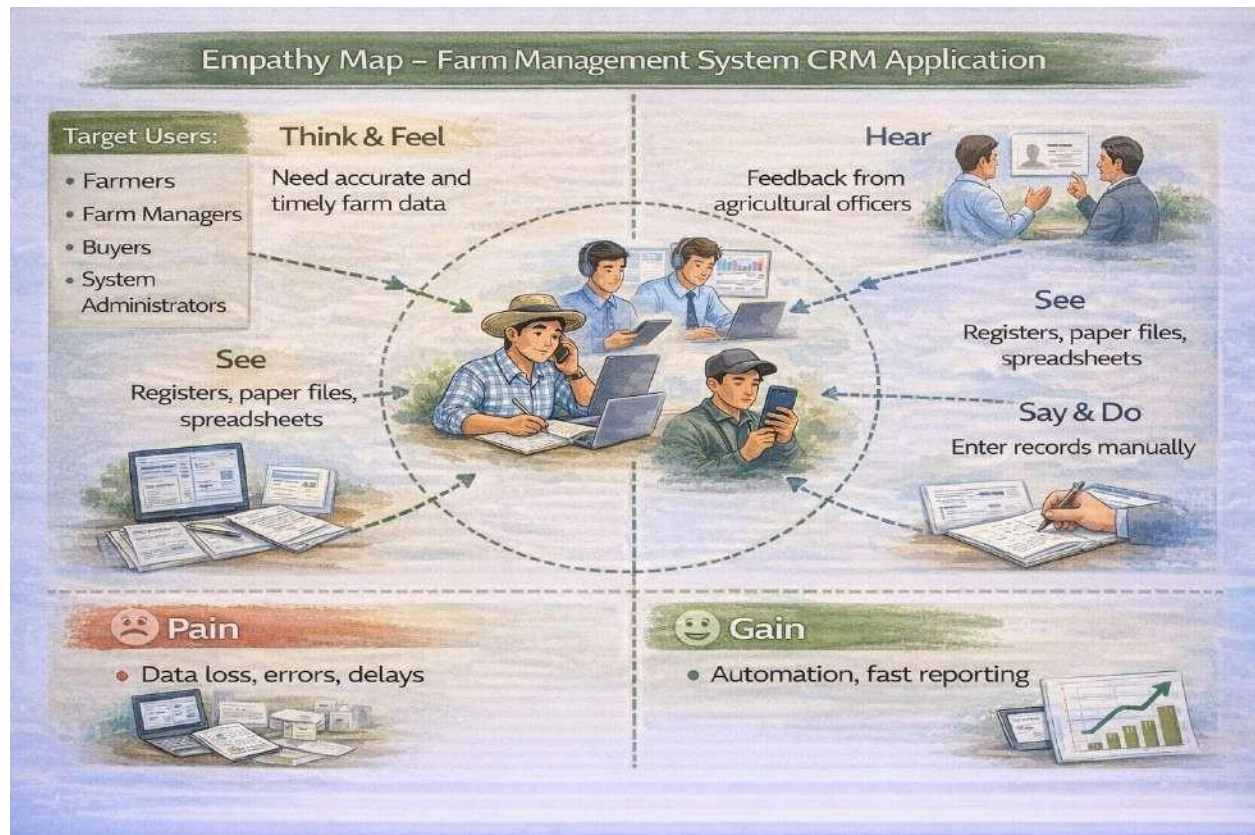
EmpathyMapCanvas

TargetUsers:

- Farmers
- FarmManagers
- Buyers
- SystemAdministrators

EmpathyAnalysis

- **Think&Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say&Do:** Enter records manually
- **Pain:** Data loss, errors, delays
- **Gain:** Automation, fast reporting



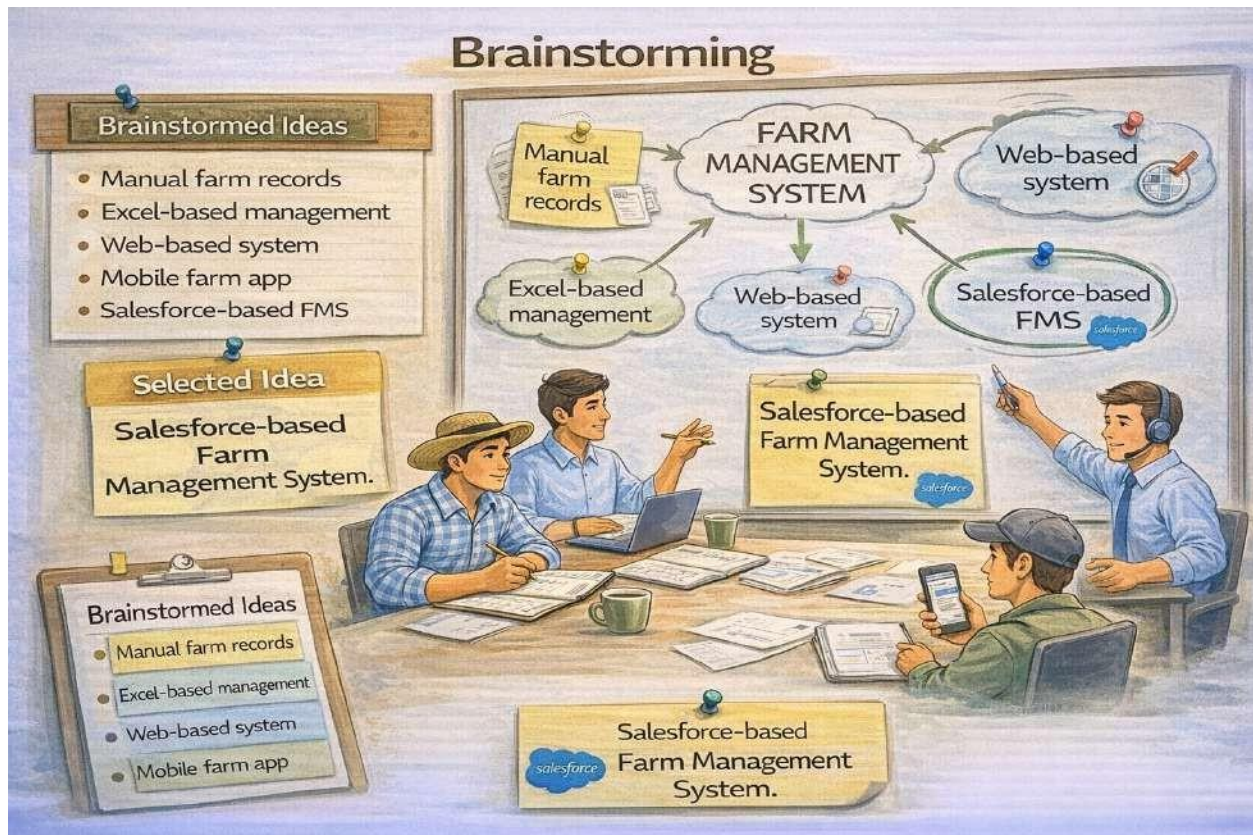
Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

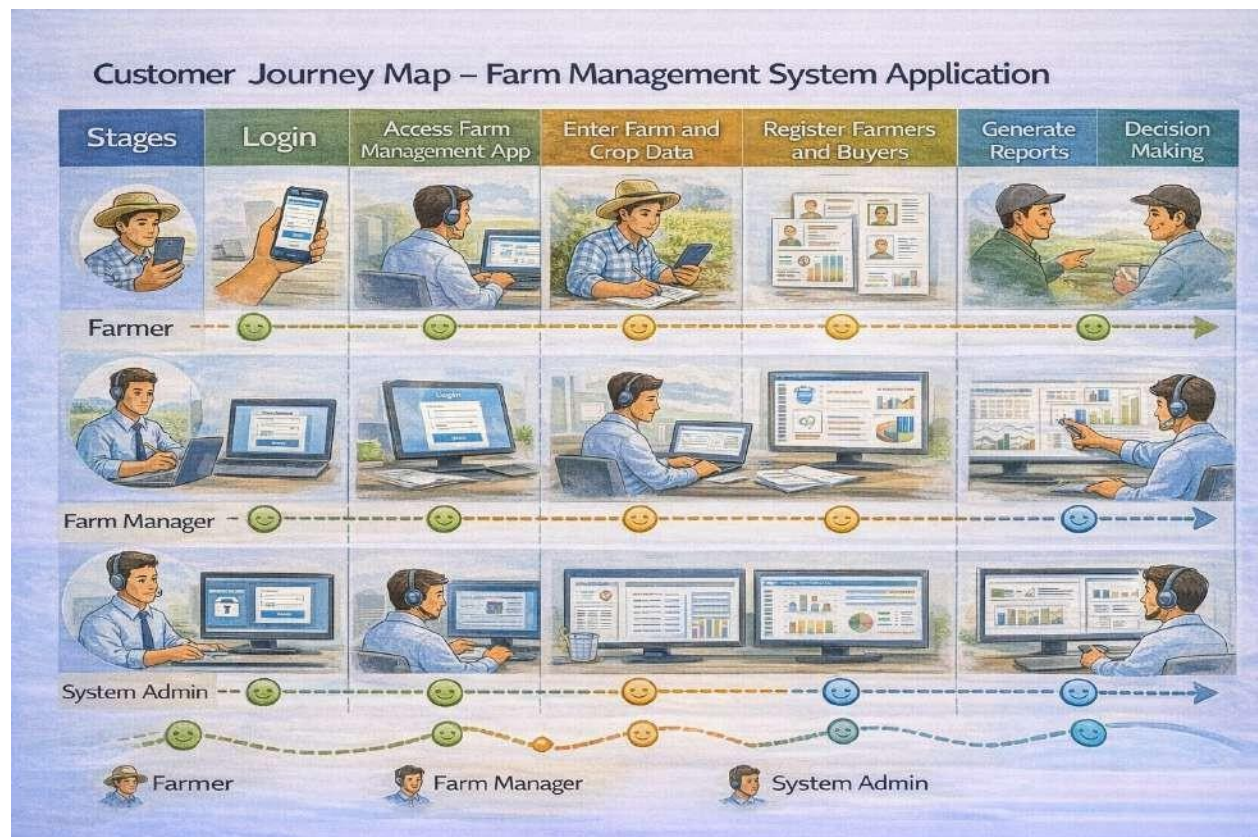


3. REQUIREMENT ANALYSIS PHASE

Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



SolutionRequirements

FunctionalRequirements

FRNo	Requirement
FR-1	UserAuthentication
FR-2	Farm,Crop,Farmer,BuyerManagement
FR-3	DataEntryandValidation
FR-4	RelationshipMapping
FR-5	ReportGeneration

FR-6	DashboardVisualization
FR-7	UserRoleManagement

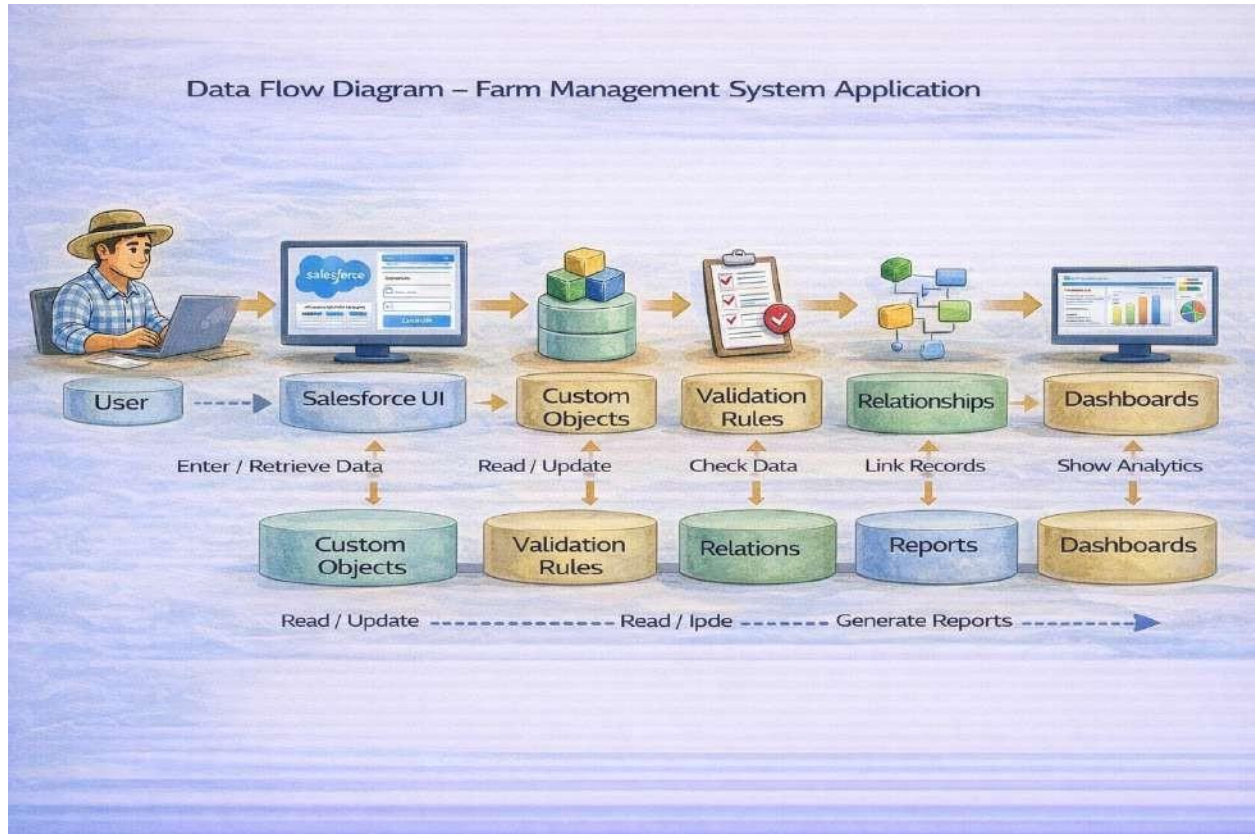
Non-FunctionalRequirements

NFR	Description	Example
Usability	Easyinterface	SimpleLightningUI
Security	Dataprotection	Profiles&roles
Reliability	Nodataloss	Cloudbackup
Performance	Fastaccess	Reportsin<3sec
Availability	24/7access	Cloudhosting
Scalability	Handlesgrowth	Morefarms/users

DataFlowDiagram

DataFlow:

User→SalesforceUI→CustomObjects(Farm, Crop, Farmer, Buyer) → Validation Rules
→ Relationships → Reports → Dashboards



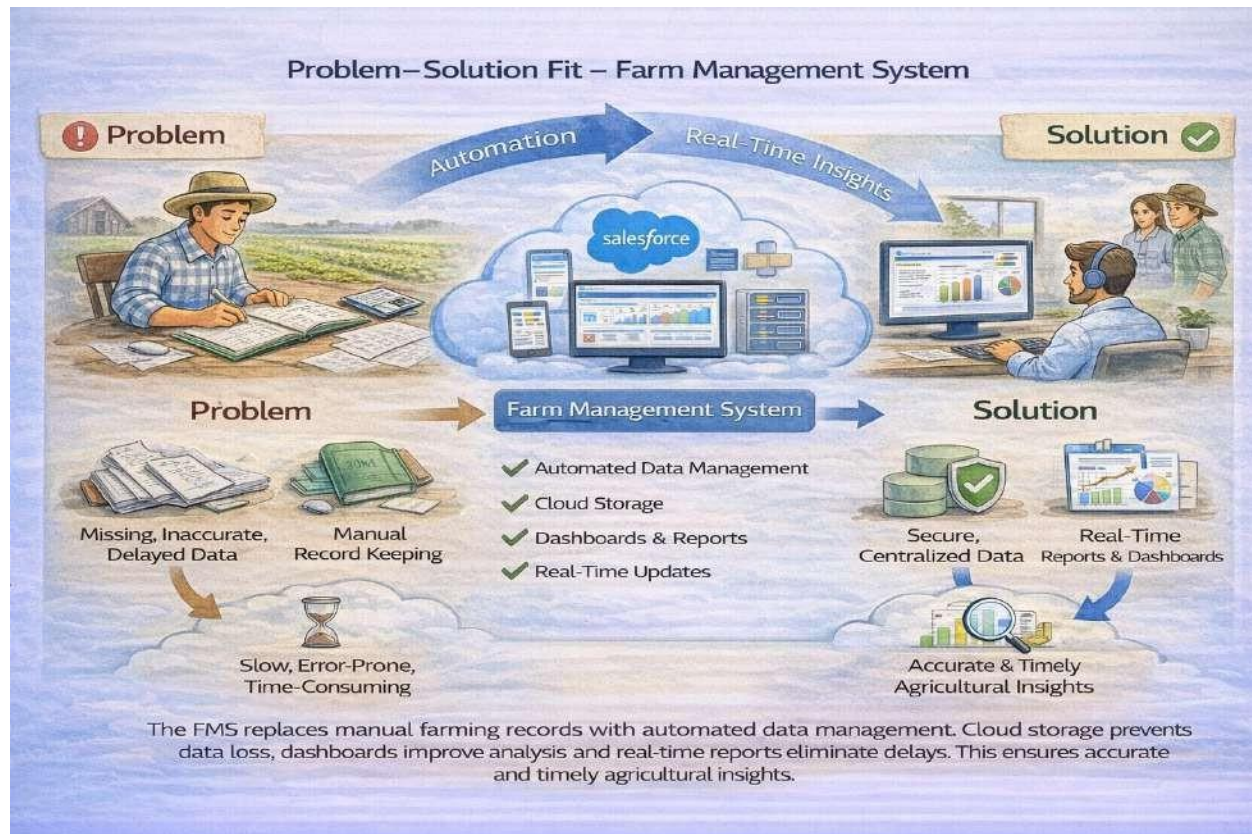
TechnologyStack

Layer	Technology
Frontend	SalesforceLightningUI
Backend	SalesforcePlatform
Database	SalesforceCloud
Automation	ValidationRules
Analytics	Reports&Dashboards

4. PROJECT DESIGN PHASE

Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



Proposed Solution

Functional Requirements

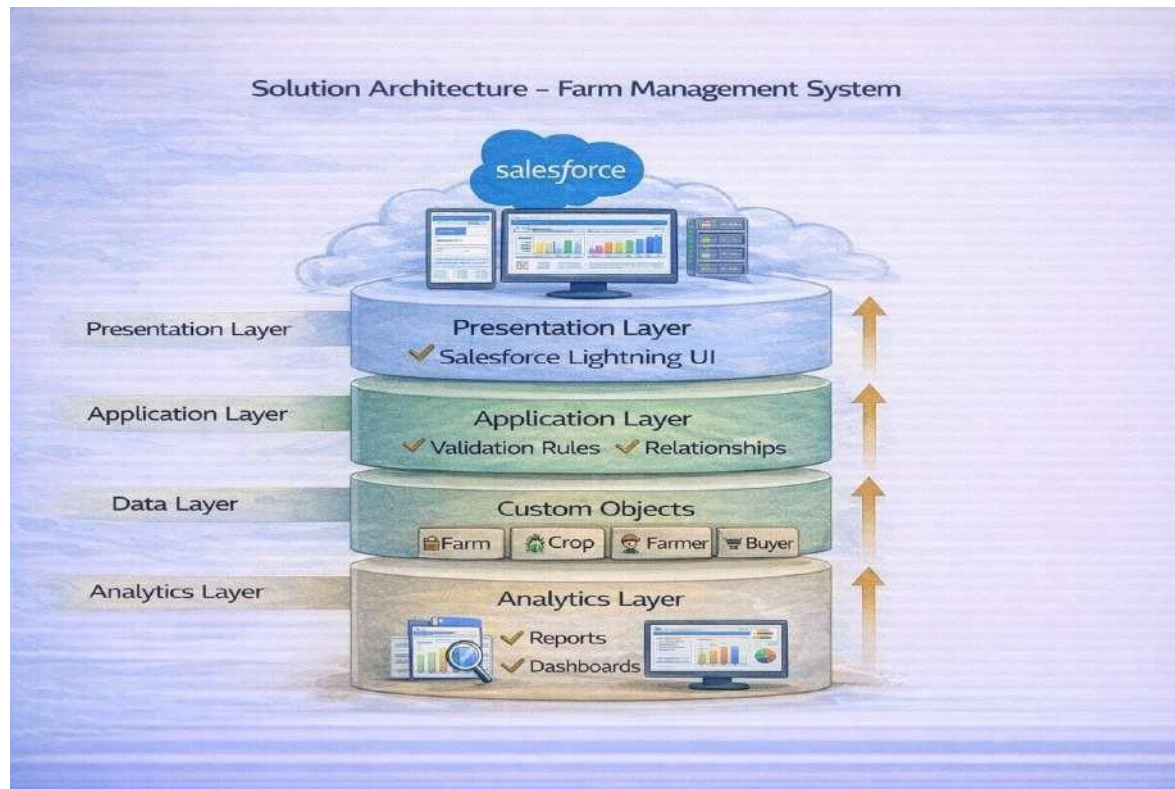
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story/Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	TotalStory Points	Duration	SprintStart Date	SprintEndDate (Planned)	StoryPoints Completed (as on PlannedEnd Date)	SprintRelease Date (Actual)
Sprint-1	20	6Days	24Oct2022	29Oct2022	20	29Oct2022
Sprint-2	20	6Days	31Oct2022	05Nov2022		
Sprint-3	20	6Days	07Nov2022	12Nov2022		
Sprint-4	20	6Days	14Nov2022	19Nov2022		

6. ProjectDevelopmentPhase:

ProjectDescription:

TheFarmManagementSystem(FMS)inSalesforceisacomprehensivesolution designedtodigitizeandstreamlineagriculturaloperations.Thisapplicationempowers farmers, farmmanagers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone1-Salesforce:

WhatIsSalesforce?

Salesforceisyourcustomersuccessplatform,designedtohelpyousell,service,market, analyze, and connect with your customers.

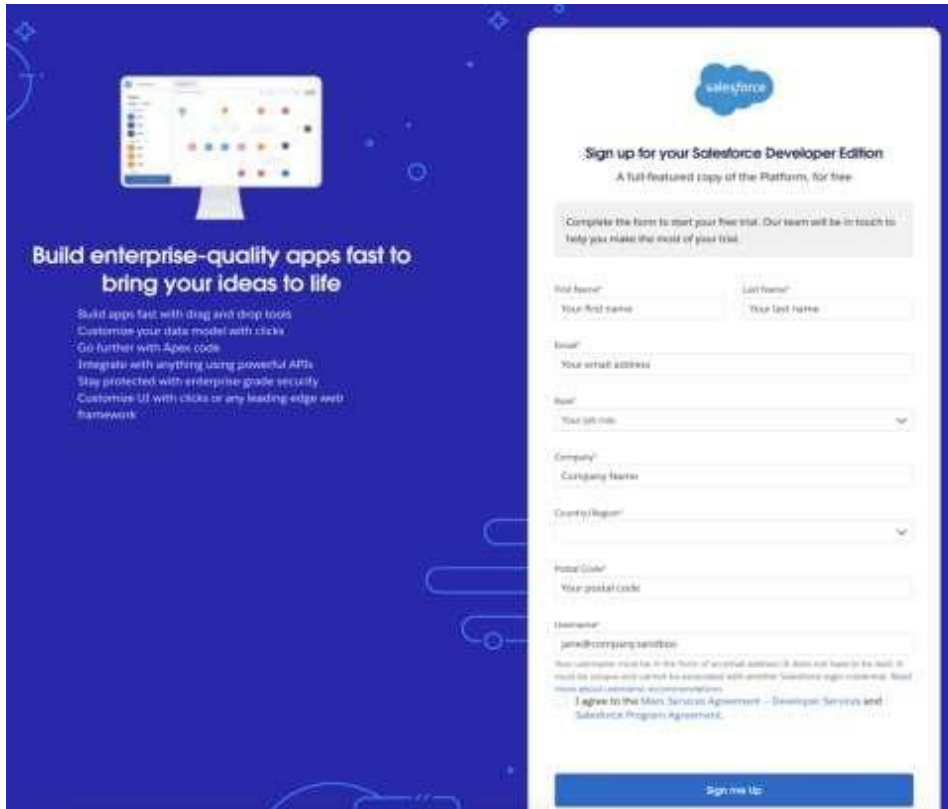
Salesforcehaseverythingyouneedtorunyourbusinessfromanywhere.Usingstandard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like Firstname, lastname, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few mins you will receive an email from salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name* Last Name*
Your first name Your last name

Email*
Your email address

Role*
Your job title

Company*
Company Name

Country/Region*

Postal Code*
Your postal code

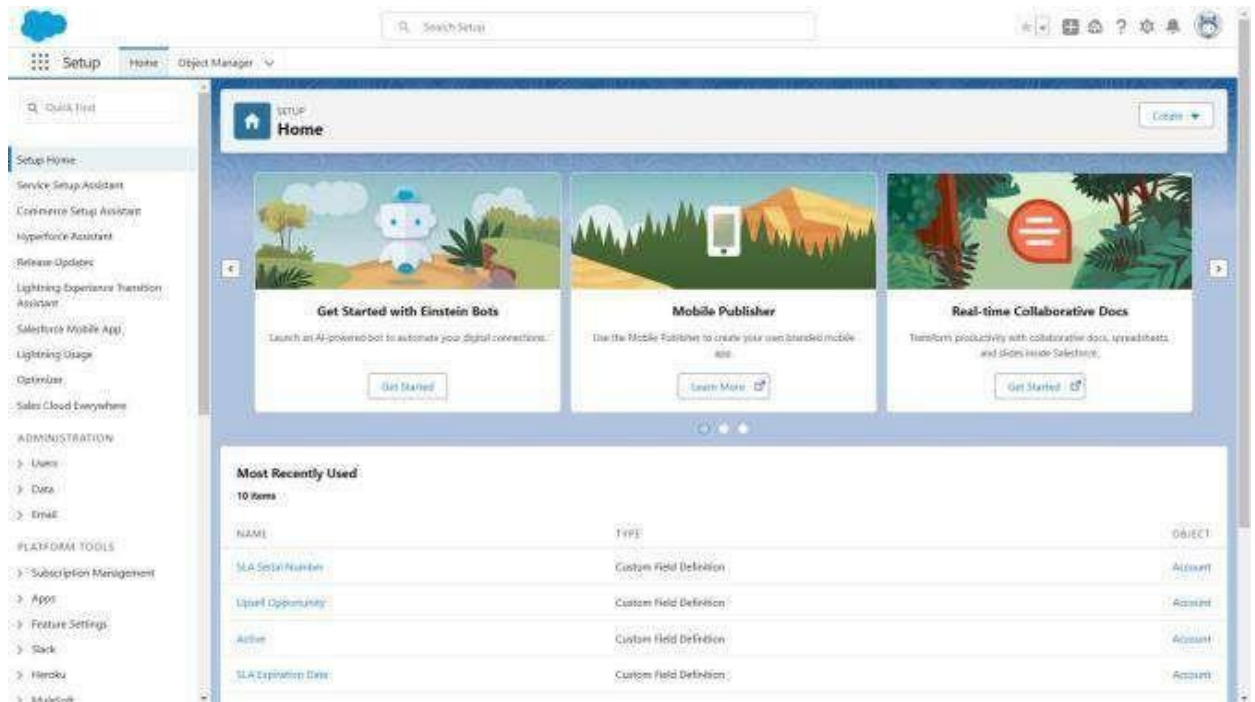
Username*
jane@company.sandbox

Your username must be in the form of an email address. It will not have to be used. It must be unique and cannot be associated with another Salesforce org instance. Read more about username requirements.

☒ I agree to the Main Service Agreement – Developer Services and Salesforce Program Agreement.

Sign me up

6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

1. Farm
2. Crop
3. Farmer
4. Buyer

Activity1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as "Farm".
4. Fill in the plural label as "Farms".
5. Record name: "FarmName".
6. Select the data type as "Text".

The screenshot shows the Salesforce Object Setup page for the 'Farm' object. The left sidebar contains a navigation menu with the following items: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Fields, Search Layouts, List View Button Layout, Restriction Rules, Sharing Rules, and Object Access. The main content area is titled 'Data Type' and shows 'Text' selected. A warning message states: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' The page is divided into several sections: 'Optional Features' with checkboxes for 'Allow Reports' (checked), 'Allow Activities' (unchecked), 'Track Field History' (unchecked), 'Allow in Chatter Groups' (unchecked), and 'Enable Lightning' (checked); 'Object Classification' with checkboxes for 'Allow Sharing' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked); 'Deployment Status' with radio buttons for 'In Development' (selected) and 'Deployed' (unchecked); and 'Search Status' with a checkbox for 'Allow Search' (checked). At the bottom right, there is a 'What's New?' link. At the bottom center, there are three buttons: 'Save', 'Save & New', and 'Cancel'.

7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: CropName
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. PluralLabel:Farmers
- d. RecordName:FarmerName e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Buyer
- c. Plural Label: Buyers
- d. Record Name: Buyer Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
- i. Click Save.

Milestone3-

Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

Setup | Home | Object Manager

Search Setup

Setup

Search tabs

User Interface

Recreate Tabs and Labels

tabs

Didn't find what you're looking for? Try using Global Search.

Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to extend external web applications and content within the Salesforce window. Visualforce tabs allow you to extend Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs

New | Refresh Table

Action	Label	Tab Style	Description
Start : New	Details	Details	
Start : New	Customization	Custom	
Start : New	Groups	Group	
Start : New	Features	Feature	
Start : New	Labels	Label	
Start : New	Job Applications	Job App	
Start : New	Skills	Skills	
Start : New	Labels	Label	
Start : New	Recruitment	Recruit	
Start : New	Students	Student	

Web Tabs

New | Refresh Table

2. Under custom object tabs, click New.

Setup

Home

Object Manager

Setup

Home

Object Manager

Search Setup

Info

Use Interface

Remove Tabs and Labels

Tab

Didn't find what you're looking for?
Try using Global Search.

SETUP

Tabs

New Custom Object Tab

Help for this Page

Step 1: Enter the Details

Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab

Get Required Information

Select an existing custom object or create a new custom object now.

Object:

Custom - v

For Style:

Custom - v

(Optional) Choose a Home Page Custom Link to show on a splash page the first time your users click on this tab.

Splash Page Custom Link:

Custom - v

Enter a short description

Description:

Next

Cancel

3. ForObject,selectFarm.

4. For TabStyle, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For TabStyle, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For TabStyle, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For TabStyle, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

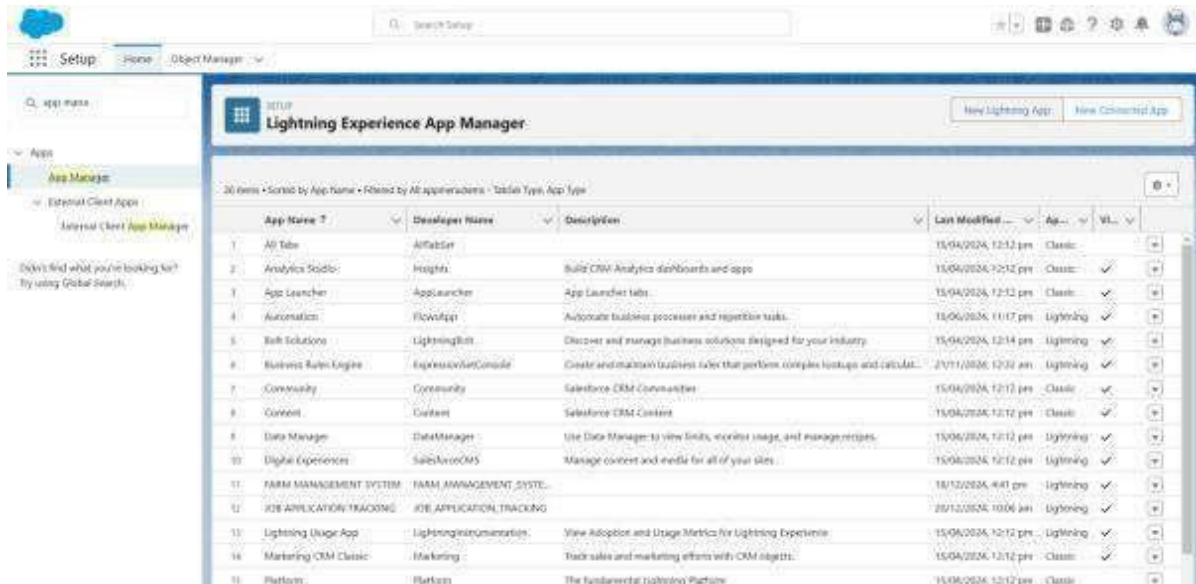
Milestone4-Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area displays a list of installed apps, sorted by App Name. The list includes various Salesforce apps such as Analytics Studio, App Launcher, Automation, Bulk Solutions, Business Rules Engine, Community, Content, Data Manager, Digital Experiences, Farm Management System, Job Application Tracking, Lightning Usage App, Marketing CRM Classic, and Platform. Each app entry shows its name, developer name, description, last modified date, app type, and version.

App Name	Developer Name	Description	Last Modified	App Type	Version
1. All Tabs	AllTabSet		15/04/2024, 12:12 pm	Classic	
2. Analytics Studio	Analytics	Build CRM Analytics dashboards and apps	15/04/2024, 12:12 pm	Classic	✓
3. App Launcher	AppLauncher	App Launcher tabs	15/04/2024, 12:12 pm	Classic	✓
4. Automation	FlowApp	Automate business processes and repetitive tasks	15/04/2024, 11:17 pm	Lightning	✓
5. Bulk Solutions	LightningBulk	Discover and manage business solutions designed for your industry	15/04/2024, 12:14 pm	Lightning	✓
6. Business Rules Engine	ExpressionEngine	Create and maintain business rules that perform complex logic and calculations	21/11/2024, 12:01 pm	Lightning	✓
7. Community	Community	Salesforce CRM Communities	15/04/2024, 12:12 pm	Classic	✓
8. Content	Content	Salesforce CRM Content	15/04/2024, 12:12 pm	Classic	✓
9. Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage ranges	15/04/2024, 12:12 pm	Lightning	✓
10. Digital Experiences	SalesforceDX	Manage content and media for all of your sites	15/04/2024, 12:12 pm	Lightning	✓
11. FARM MANAGEMENT SYSTEM	FARM MANAGEMENT SYSTEM		18/12/2024, 4:41 pm	Lightning	✓
12. JOB APPLICATION TRACKING	JOB APPLICATION TRACKING		20/12/2024, 10:06 am	Lightning	✓
13. Lightning Usage App	LightningUsageApp	View Adoption and Usage Metrics for Lightning Experience	15/04/2024, 12:12 pm	Lightning	✓
14. Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects	15/04/2024, 12:12 pm	Classic	✓
15. Platform	Platform	The fundamental Lightning Platform	15/04/2024, 12:12 pm	Classic	

- From Setup, enter App Manager in the Quick Find and select App Manager.

- Click New Lightning App. Farm Management System as the App Name, then click Next

- Under App Options, leave the default selections and click Next.

- Under Utility Items, leave as is and click Next.

- From Available Items, select Farm, Crop, Farmer, Buyer

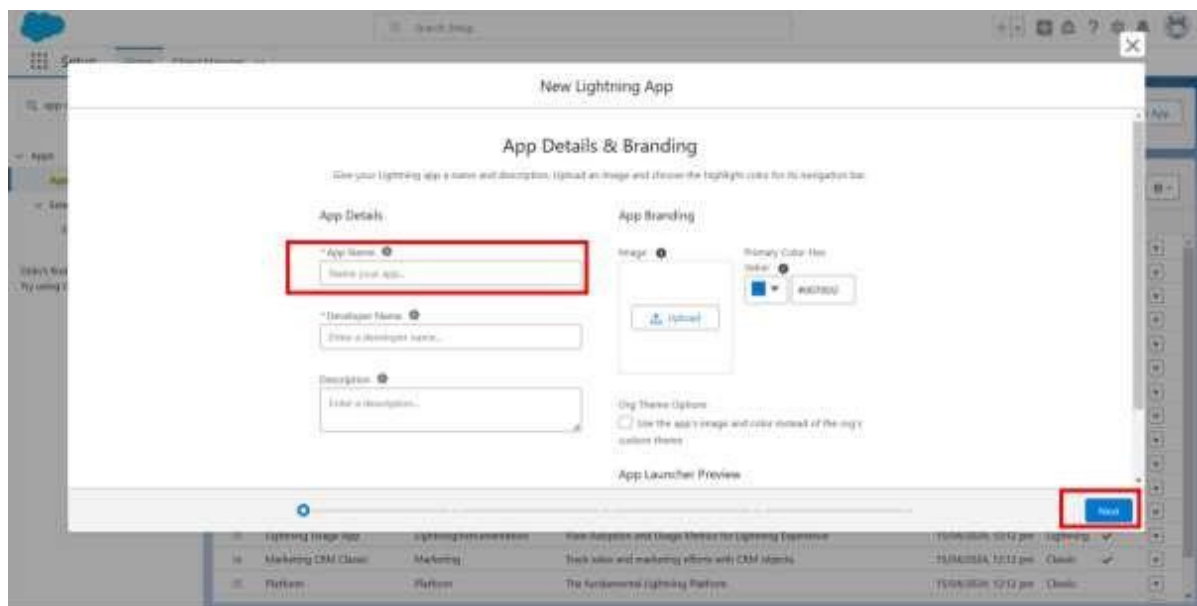
Reports, and Dashboards and move them to Selected Items. Click Next. ● From Available Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

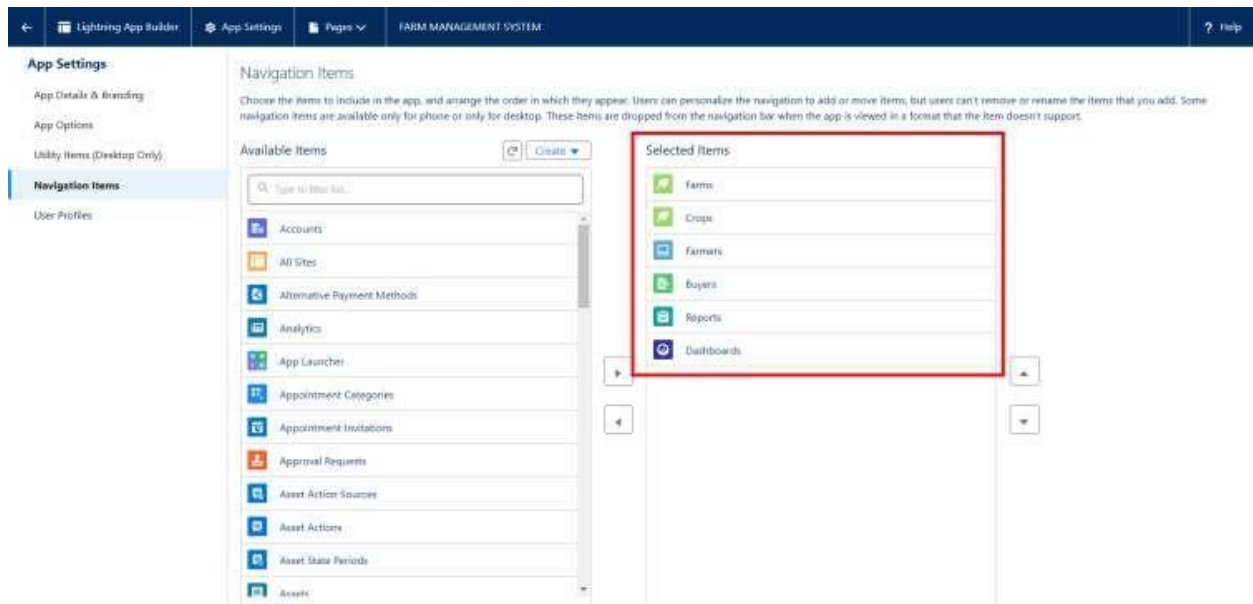
Note:

1. App Launcher-Displays available apps.

2. App Name-Displays the current selected app.



3. Navigation menu-Displays the tabs available inside the app.



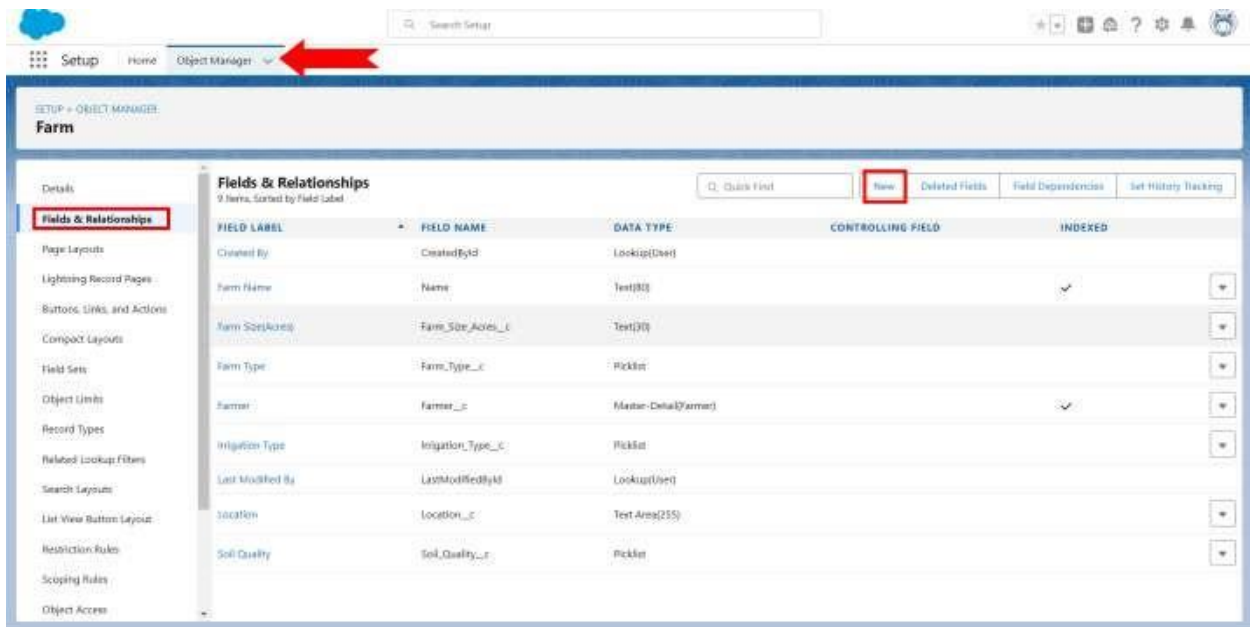
Milestone5- FieldsandRelationship:Whatarefields?

FieldsinSalesforcerepresentwhatthecolumnsrepresentinrelationaldatabases.Itcan store data values which are required for a particular object in a record.

Activity1:

CreationoffieldsfortheFarmobjects:

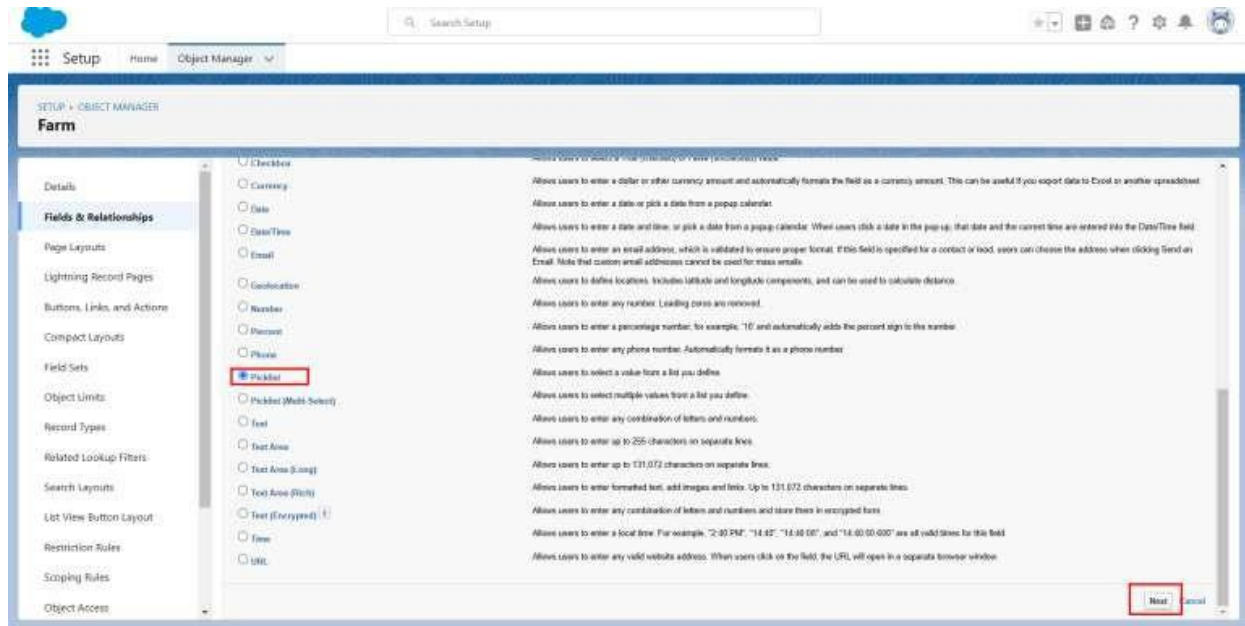
1. clickthegeariconandselectSetup.ThislaunchesSetupin anew tab.
2. ClicktheObjectManagertabnexttoHome.
3. SelectFarm.
4. SelectFields&Relationshipsfromthelleftnavigation,andclick New



The screenshot shows the Salesforce Setup page for the Farm object. The top navigation bar includes tabs for Setup, Home, and Object Manager. A red arrow points to the Object Manager tab. The left sidebar shows the navigation menu with 'Fields & Relationships' highlighted. The main content area displays the 'Fields & Relationships' section for the Farm object, showing a table of fields.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Name	Name	Text(80)		✓
Farm Size (Acres)	Farm_Size_Acres__c	Text(30)		
Farm Type	Farm_Type__c	Picklist		
Farmer	Farmer__c	Master-Detail(Farmer)		✓
Irrigation Type	Irrigation_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	Text Area(255)		
Soil Quality	Soil_Quality__c	Picklist		

5. Choose Datatype as a picklist, Click next



6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
7. Next, Next Click Save.

Setup Home Object Manager

Search Setup

SETUP > OBJECT MANAGER

Farm

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

New Custom Field

Step 2: Enter the details

Step 2 of 4

Field Label: Farm Type

Values:

- ☐ Use global picklist value set
- ☒ Enter values, with each value separated by a new line

Crop
Livestock
Mixed

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field Name: Farm_Type

Description:

Help Text:

Now let's create the other fields following Activity 1 steps 1 to 4 and we must choose the data types of the fields carefully.

1. Next we create another picklist field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
2. Again we create picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)
3. Choose data type as text, enter field label (Location)
4. Choose data type as text, enter field label (size)
5. Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

1. Choose Data type as text, Enter Field Label name (Crop Name), Next save.
2. Select Data type as date, Enter Field label (Planting Date), then save.
3. Again we select data type as date, Enter field label (Harvest date), save
4. Next we create lookup relationship on crop object related to Farm Object.
5. We select data type as currency, Enter field label (Actual Yield), Next Save it.
6. Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length (20) and save.
7. Now we create formula field for that select data type as a formula, Click Next.

Crop

Chapelle

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layout

Field Sets

Object Units

Material Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scirpus flaccus

Object Across

Specify the type of information that the custom field will contain:

Data Type

 Springer

Select one of the data types below

 Add New

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

45/50

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

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A read-only field that displays the sum, minimum, or maximum value of a field in a related set or the record count of all records listed in a related set.

Looking Relatively

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

 [More Detail Information](#)

Creates a special type of parent-child relationship between this object (the child, or 'leaf') and another object (the parent, or 'master') where

- The relationship field is required for all detail records.
 - The ownership and tracking of a detail record are determined by the master record.
 - If there is a one-to-one relationship, all detail records are deleted.
 - You can create rollup summary fields on the master record to summarize the detail records.
- The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.

The `relationship` field allows users to click on a linkup icon to select a value from a popup list. The master object is the source of the values in the list.

○ External Learning Relationship

Create a relationship that links this object to an external object whose data is stored outside the Salesforce org.

Objective

Allow users to select a True (checked) or False (unchecked) value

 Currency

Always want to enter a dollar or other currency amount and automatically format the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.

Criteria

Allows users to enter a date or pick a date from a group calendar

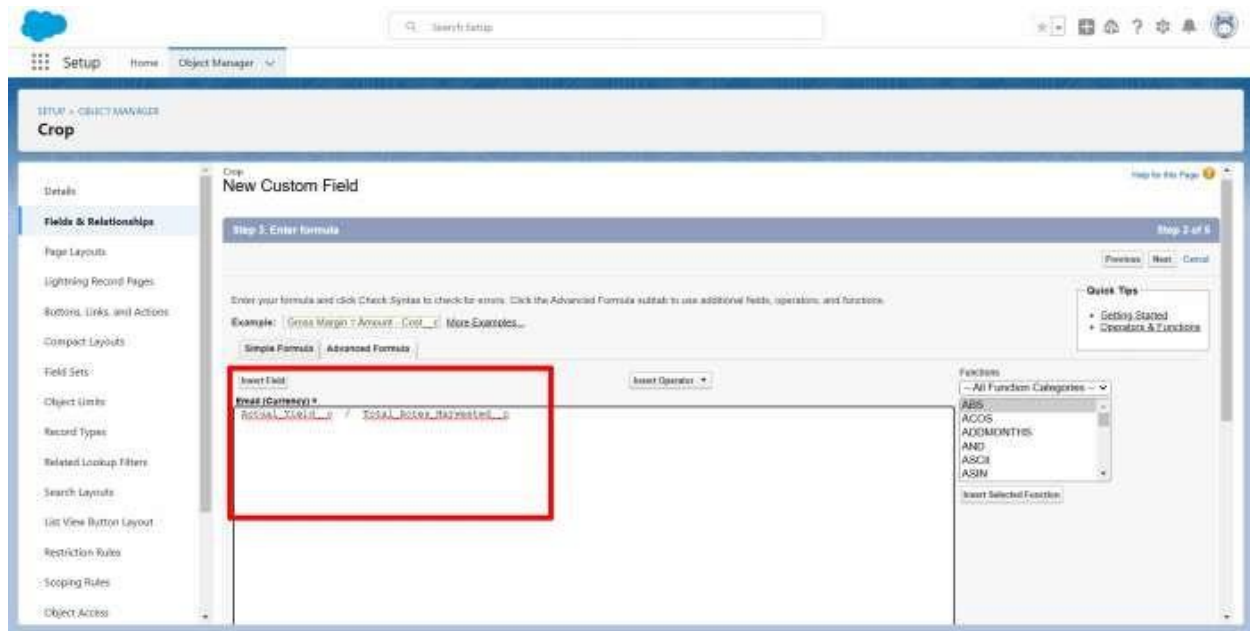
100

Journal of Management Education 35(10) 1109-1124

Email: Note that custom email addresses cannot be used for mass emails.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity3:

CreationoffieldsfortheFarmerobjects:

1. SelectDatatypePhone, Enterfieldlabel(ContactNumber), nextSave.
2. SelectDatatypePicklist, EnterFieldLabel(State), Enter values(Telangana, Andhra Pradesh, Tamilnadu), save.
3. ChooseDatatypePicklist, EnterFieldLabel(District), Enter Valuesofdistrict, nextsave.
4. AgainselectDatatypeasPicklist, EnterFieldLabel(Village), Enter Valuesof villages, next save.

Creationoffielddependency:

1. NowcreateFieldDependencyForDistrictfield, OpenObjectManager, ChooseFarmer object ,select Fields&Relationships, On Right side We have Field Dependency option, select that option.
2. Clicknew, SelectControllingFieldasState, DependentFieldAsDistrict, Clickon Continue.
3. SelecttheappropriateValueineachcolumnbydouble-clickingthem.
4. ForEx. Telangana- YadadriBhongir.
5. ClickIncludeValues. AnditisalsothesameforAndhraPradesh, Tamilnaduwithits city.
6. ClickPreview, thentestthependencybyselectingadifferentStateandviewingthe associate Values available for Particular state.



Search Setup



Setup

Home

Object Manager

SETUP • OBJECT MANAGER

Farmer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

New Field Dependency

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that does filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 200 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1. Select a controlling field and a dependent field. Click Continue when finished.

Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling Field

Dependent Field

None

None

Continue

Cancel

Continue

Cancel

7. Click Close to close the preview window, Click Save.

Activity4:

Creation of fields for the Buyer objects:

1. Select Datatype as a Phone, Enter a Field Label (Contact), Save.
2. Select Datatype as an Email, Enter Field Label (Email), Next save.
3. Select Datatype Textarea, Enter Field Label (Address), Save it.
4. Select Datatype Picklist, Field Label (Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone6-Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges. A profile can be assigned to many users, but a user can be assigned a single profile at a time.

Activity 1:

Create a custom profile:

1. From setup, enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

Setup

Home

Object Manager

Search Setup

profile

Users

Profiles

Didn't find what you're looking for?

Try using Global Search.

Setup

Profiles

Help for this Page

All Profiles

201 (2 items)

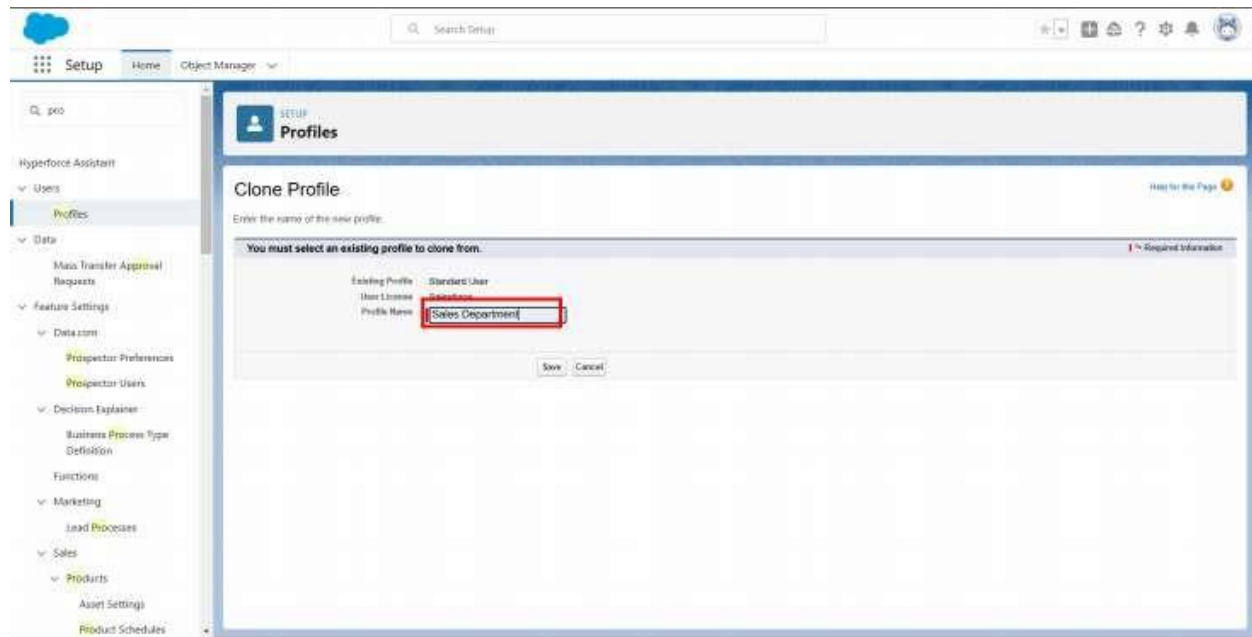
Create new view

New Profile

1 2

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other

☐	Action	Profile Name	User License	Custom
☐	Edit (1)...	Sales Development	Salesforce Platform	☒
☐	Edit (1)...	Sales Manager	Salesforce	☒
☐	Edit (1)...	Salesforce App, Sales System Administrator	Salesforce Integration	☒
☐	Edit (1)...	Sales Partner User	Sales Partner	☐
☐	Edit (1)...	Salesforce Manager	Salesforce	☐
☐	Edit (1)...	Salesforce Platform User	Salesforce Platform	☐
☐	Edit (1)...	Salesforce Platform User	Salesforce Platform	☒
☐	Edit (1)...	Salesforce User	Salesforce	☐
☐	Edit (1)...	System Administrator	Salesforce	☐



5. Clicksave.

Activity2:

Create a custom profile-2:

Create a profile with the profile name as “FarmManager”.

6. From setup, enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

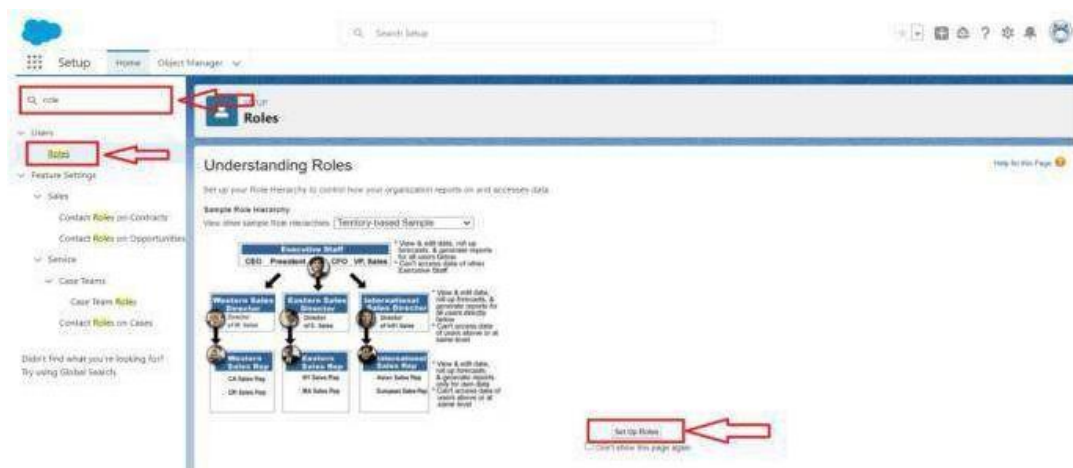
Milestone7-Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

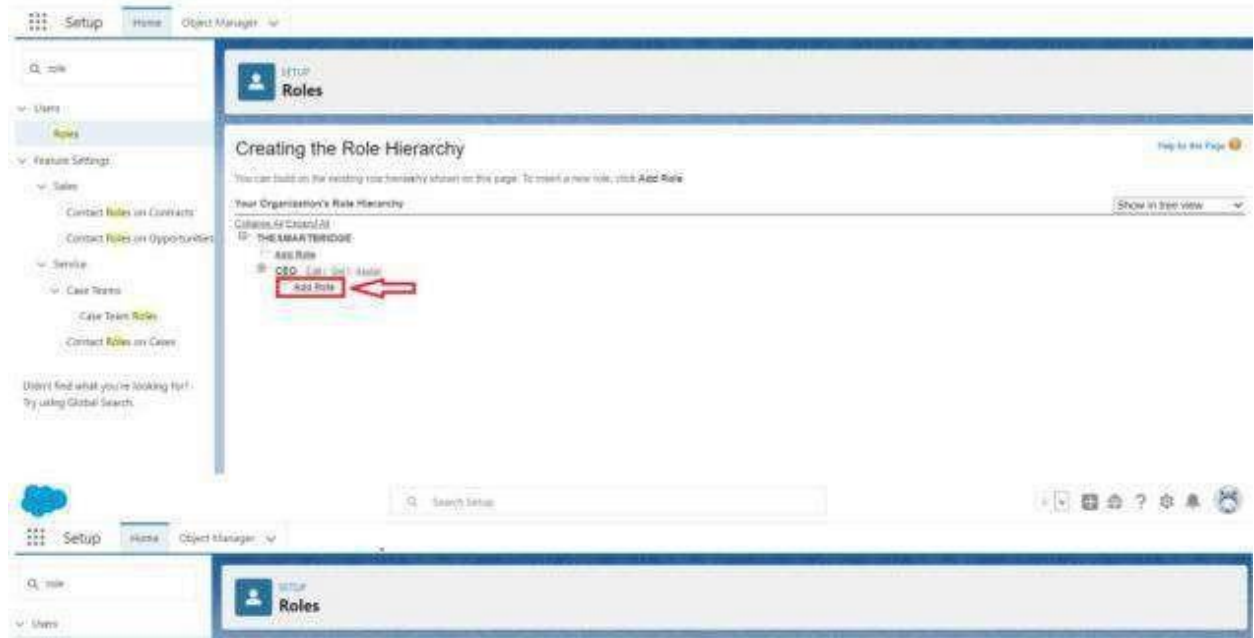
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the setup roles option



3. Below the CEO click on add role and enter the label name as "FarmManager" and role name will be Automatically populated and click on save.



Milestone8-User:

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE-As Salesforce license can only be used by 2 Users at a time in Dev Org, so if you don't find Salesforce license then deactivate a user who has a Salesforce license or change the license type from Salesforce to any other.

Activity1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup interface. In the left sidebar, the 'Users' link is highlighted with a red box. The main content area displays the 'All Users' page. At the top of this page, the 'New User' button is highlighted with a red box. Below the button, there is a table of users with columns: Active, Full Name, Alias, Username, Role, and Profile. The table contains several rows of user data, including usernames like 'jdoe', 'jdoe2', 'jdoe3', etc., and roles like 'Standard User', 'System Administrator', etc.

Active	Full Name	Alias	Username	Role	Profile
☐	J. Doe	jdoe	jdoe@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe2	jdoe2@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe3	jdoe3@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe4	jdoe4@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe5	jdoe5@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe6	jdoe6@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe7	jdoe7@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe8	jdoe8@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe9	jdoe9@salesforce.com	STU	Standard Platform User
☐	J. Doe	jdoe10	jdoe10@salesforce.com	STU	Standard Platform User

3. Enter Firstname as Far and lastname as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address.
By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile (Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile.
Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Go to Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with this section select roles and in that select Farm Manager.
- 8) And in this section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

Setup

Profile

Object Manager

Q sharing

Security

Group User Sharing Rule Access Report

Sharing Settings

Didn't find what you're looking for?
Try using Global Search.

Sharing Settings

Use sharing rules to make automatic decisions to your organization-wide sharing settings for defined sets of users.
Note: "Roles and subordinates" includes all users in a role, and the roles below that role.
You can use sharing rules only to grant access; you can't deny or restrict access.

Label Crop

Rule Name Crop

Description

Step 1: Select your rule type

Criteria

Field	Operator	Value	
Crop Name	Equals	True	AND
None	None	None	AND
None	None	None	AND
None	None	None	AND
None	None	None	AND

Include Records

Include records owned by users who can't have an assigned role

Start with Rule From Manager

Access Level Read/Write

Created By Salsalatt, Jattala 10/12/2024 2:20 pm

Modified By Salsalatt, Jattala 10/12/2024 2:20 pm

Save Cancel

Activity2:

Create a Sharing Rule to Share the records of Buy to Sales department with the Access of Read/Write.
Follow the steps from above Activity.

Milestone-10: User Adoption

Activity1: Create a Record (Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows the 'Edit Kesava' form in the Farm Management app. The form is titled 'Edit Kesava' and has a 'Required Information' label. The fields are as follows:

- Farmer Name: Kesava
- Contact Number: 98260779
- State: Andhra Pradesh
- District: Eluru
- Village: Venkatesa
- Post (Pincode): 15

The 'Save' button is highlighted with a red box. The background shows the 'FARM MANAGEMENT' app interface with a 'Recently Viewed' list and a 'Farmer Name' tab.

Milestone 11-Reports:

What are Reports?

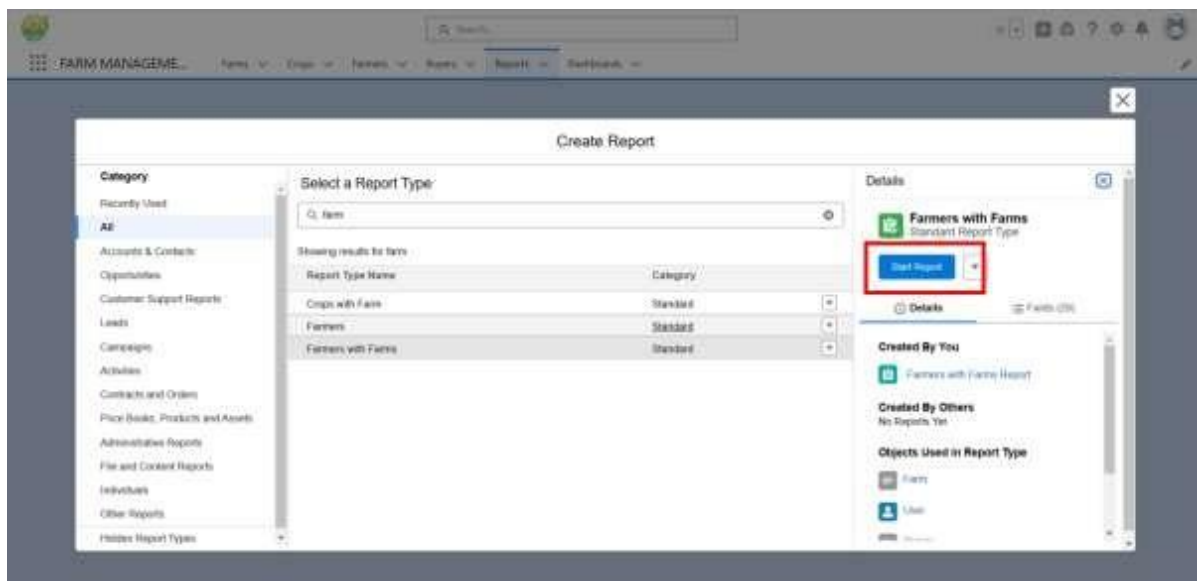
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group, Select Which Fields You Want to Group.
- 6) Save the report by giving label name (Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot displays the 'FARM MANAGER' application interface, specifically the 'REPORTS' section. The main title is 'Farmers with Farms Report'. The interface includes a search bar at the top, navigation tabs for 'Farms', 'Crops', 'Farmers', 'Reports', and 'Dashboard'. The 'REPORTS' tab is active, showing a list of reports. The selected report is 'Farmers with Farms Report', which is currently in 'Preview' mode. The report data is displayed in a table with the following columns: 'Irrigation Type', 'Farmer: Farmer Name', 'Farm: Farm Name', 'Contact Number', 'Farm Size(Acres)', and 'Farm Type'. The table contains three data rows and a 'Total' row. The 'Irrigation Type' column has a dropdown menu set to 'Localized Irrigation (3)'. The 'Farm Type' column has a dropdown menu set to 'Livestock'. The 'Total' row shows a sum of 25 for 'Farm Size(Acres)'. The interface also includes a 'Columns' section on the left with a list of fields to be displayed in the report, and a 'Groups' section with a list of fields to be grouped. The 'Columns' section includes 'Farmer: Farmer Name', 'Farm: Farm Name', 'Contact Number', 'Farm Size(Acres)', and 'Farm Type'. The 'Groups' section includes 'Irrigation Type'. The interface also features a 'Row Counts' section with a table showing counts for 'Row Counts', 'Detail Rows', 'Subtotal', and 'Grand Total'. The 'Row Counts' table has a 'Row Counts' column and a 'Count' column. The 'Detail Rows' column has a value of 25. The 'Subtotal' column has a value of 25. The 'Grand Total' column has a value of 25. The interface also includes a 'Conditional Formatting' button at the bottom right.

Irrigation Type	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Localized Irrigation (3)	Rajesh	Paddy Farm	989430779	10	Livestock
	Rajesh	Rajesh Farm	989430779	10	Crop
	Rajesh	Wheat Farm	788202779	15	Livestock
Total (3)				25	

Activity2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

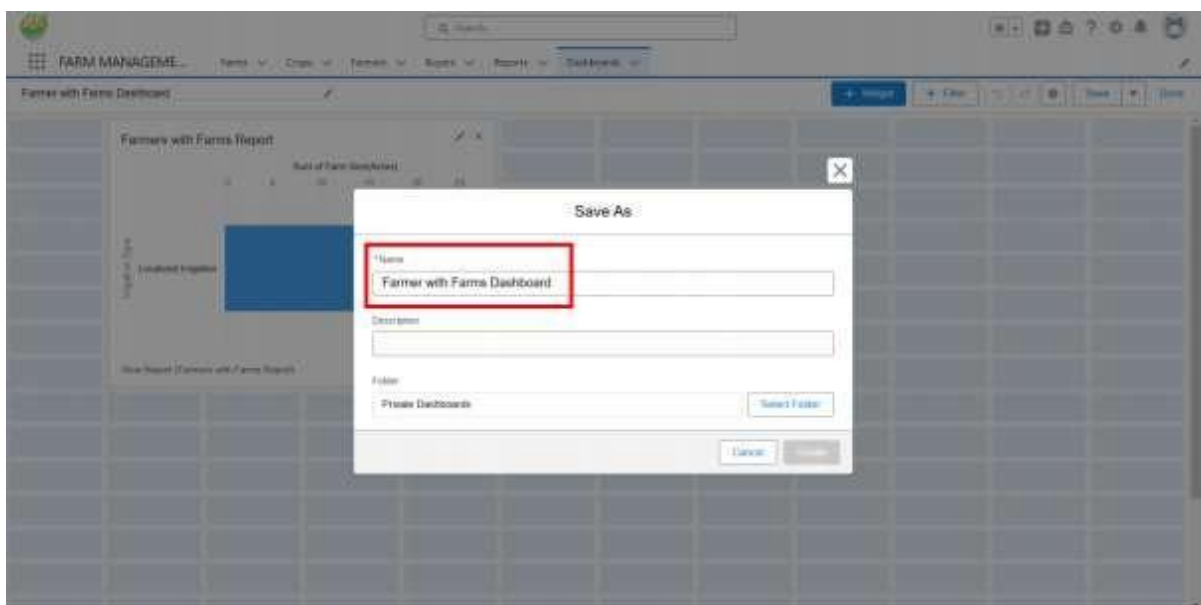
Milestone12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity1:

Create a Dashboard:

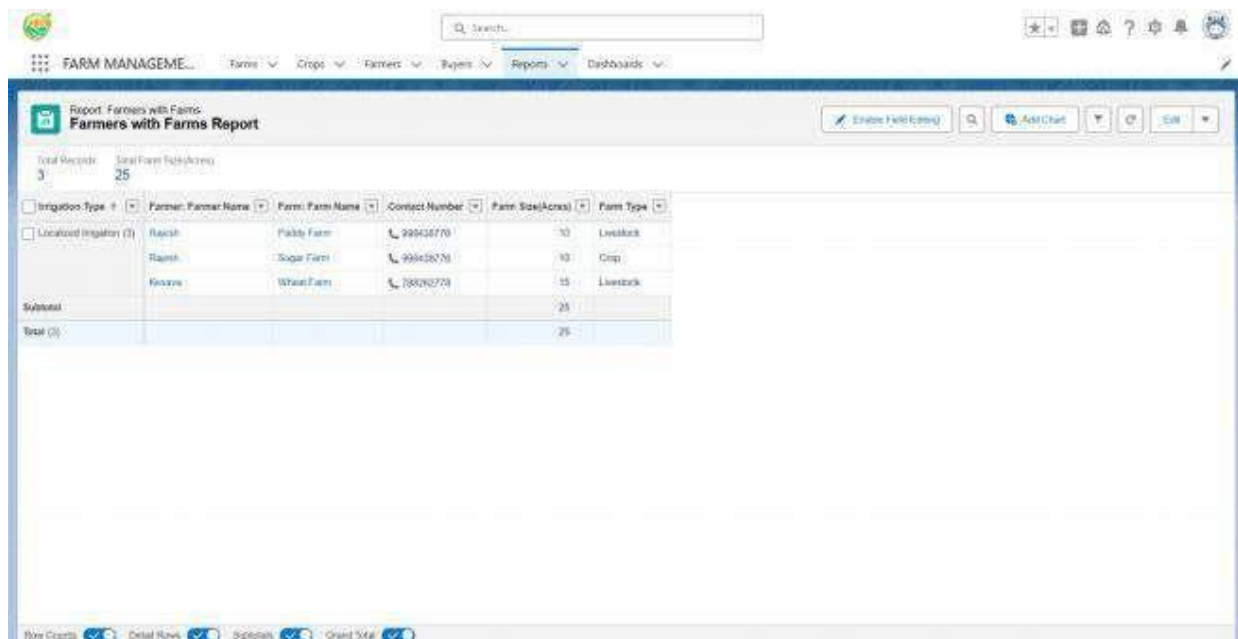
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click + Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13:ViewReportandDashboard: Activity

1: Report

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementSystem&clickonit.
- ClickonReports Tab.
- ClickonFarmerswithFarms&seerecords



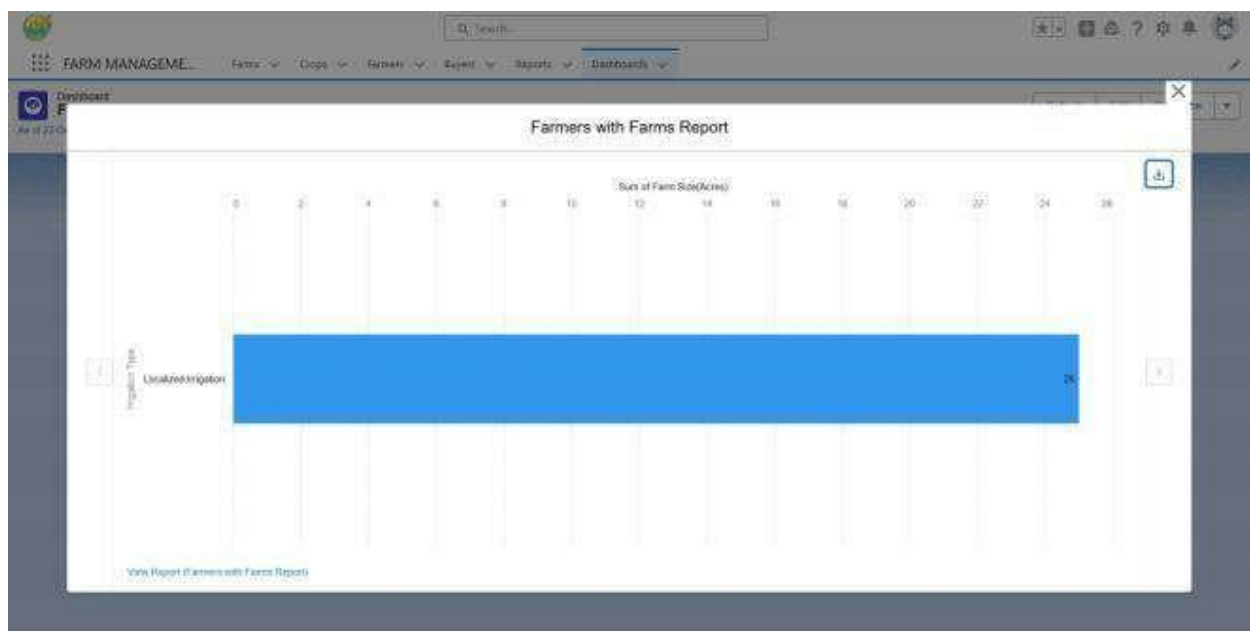
The screenshot displays the 'Farmers with Farms Report' in a web application. The interface includes a top navigation bar with tabs for Farms, Crops, Farmers, Buyers, Reports, and Dashboard. The 'Reports' tab is active. Below the navigation bar, there's a search bar and a 'Show Field Editor' button. The report title is 'Report: Farmers with Farms: Farmers with Farms Report'. It shows 'Total Records: 3' and 'Total Farm Size(Acres): 25'. The table below lists the records:

☐ Irrigation Type	☐ Localized Irrigation (3)	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
		Rajesh	Paddy Farm	999010770	10	Livestock
		Rajesh	Sugar Farm	999010770	10	Crop
		Raviya	Wheat Farm	988062728	15	Livestock
Subtotal					25	
Total (3)					25	

At the bottom, there are toggle switches for 'Show Counts', 'Detail View', 'Subtotals', and 'Grand Total', all of which are currently turned on.

Activity2:Dashboard

- ClickonAppLauncheronleftsideofscreen.
- SearchFarmManagementsystem&clickonit.
- ClickonDashboardTab.
- ClickFarmersWithfarms&seerecords.



7. FUNCTIONAL&PERFORMANCETESTING

Duringthetestingphase,screenshotswerecapturedfromtheSalesforceinterfacetoverify system functionality. The following were tested:

- CreationofFarm,Crop,Farmer,andBuyerrecords
- Validationofinputfields
- Relationshipanddependencychecks
- Profileandroleaccesstesting
- Sharingruleverification
- Reportanddashboardgeneration

Performancetestingensuredsmoothsystemresponseandstabilityduringmultipleuser operations.

8. ADVANTAGES& DISADVANTAGES

Advantages

- Centralizeddatasystem
- Reducedpaperwork
- Improvedaccuracy
- Bettermonitoring
- Real-timeanalytics

Disadvantages

- Requiresinternetconnection
- Salesforcelicensedependency
- Limitedofflineaccess

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURESCOPE

- Mobileapplicationintegration
- AI-basedcropprediction
- Weatherdataintegration
- IoTsensorconnectivity
- Governmentschemeintegration
- Marketpriceforecasting

11. APPENDIX

- SourceCode:SalesforceConfigurationandMetadata
- Dataset:Farm,Crop,Farmer,andBuyerRecords
- ToolsUsed:SalesforceDeveloperOrg

Thank You

